Week 3 – Workshop
Exploring & Understanding Inversion

Adelaide University

Geophysics 3A: Potential Fields & Geothermics

July 6, 2026

Duration: 3 hours **Setting:** Lecture room with laptops (sphere inversion app)

Preparation

Pre-reading:
Lecture videos:
Notes:

Purpose

This workshop introduces the structure of inverse problems in potential-field geophysics, contrasting forward prediction with inverse estimation. Students explore the role of misfit, data weighting, and regularization, and examine how non-linearity and non-uniqueness arise in gravity modelling. Through a sequence of activities, they classify linear vs. non-linear models, reason qualitatively about non-uniqueness, and engage in hands-on inversion of a buried sphere using grid search and Gauss–Newton methods.

Learning Outcomes

- Distinguish forward and inverse problems and articulate the role of misfit, weighting, and regularization.
- Classify simple geophysical relations as linear or non-linear and propose appropriate linearizations.
- Recognize manifestations of non-uniqueness in gravity and magnetic profiles and propose discriminating data.
- Derive or confirm analytical Jacobian expressions for the buried sphere model.
- Interpret misfit surfaces, inversion paths, and the effect of noise and damping.
- Explain parameter trade-offs and resolution limitations inherent to the buried-sphere problem.

Reference concepts align with Week 3 notes on forward/inverse modelling, misfit structure, and non-linear least squares.

Roles and Grouping

Create mixed groups (4–5 students). Recommended roles: *Facilitator*, *Analyst*, *Recorder*, *Skeptic*, *Reporter*. Rotate roles between major activities.

Session Flow (180 min)

0–30 min: Mini-lecture — Forward vs Inverse, Misfit, Weighting, Challenges

- Forward vs. inverse models: $m \mapsto g(m)$ vs. $d \mapsto m$.
- Misfit: residual $r = d - g(m)$; weighted least squares $\Phi_d = \|W_d r\|_2^2$.

- Role of W_d (noise) and W_m (model constraints); damping λ.
- Challenges: non-linearity (Jacobian needed), non-uniqueness (multiple m give similar $g(m)$), ill-conditioning.

30–60 min: Activity A: Linear or Non-linear?

Using the equation handout:

- Identify model parameters and classify each relation as *linear*, *non-linear*, or *linearizable*.
- If non-linear, propose a linearization (log-transform, re-parameterization, Taylor expansion/Jacobian).
- Deliverable: brief notes per group.

70–90 min: Non-uniqueness (Jigsaw)

Curves-only reasoning:

- Scenario A: gravity profiles — two distinct geometric models can reproduce similar anomalies.
- Scenario B: magnetic dike profiles — width/depth/magnetization trade-offs.
- For *each* curve: propose at least two plausible geologic/geometric models.
- List discriminating data (station density, gradients, susceptibility/remanence, multi-component data, mapping).
- Deliverable: 1-paragraph argument explaining which model to pursue first and what data to collect next.

90–100 min: Buried Sphere — Introduction

- App overview: Data & Model view; Misfit view; controls for acquisition, noise, synthetic model.
- Construct a true model & generate data (freeze a single noise realization).

100–120 min: Buried Sphere — Jacobian Derivatives

1. Students derive/confirm analytical expressions for $\partial g/\partial z$, $\partial g/\partial R$, $\partial g/\partial \Delta\rho$.
2. Compare with the app's Jacobian; discuss relative sensitivities: amplitude $\propto R^3\Delta\rho$, width controlled by depth z.

120–155 min: Buried Sphere — Investigation

- **Grid search:** Explore pairwise misfit slices and 3-D misfit surface. Identify local vs global minima.
- **Inversion path:** Run Gauss–Newton with different λ; examine step length and stability.
- **Effect of noise:** Increase σ; observe widening misfit basins and parameter uncertainty.
- Discuss links to curvature, conditioning, and resolution.

155–170 min: Buried Sphere — Non-uniqueness

- Demonstrate parameter trade-offs (e.g. $R^3\Delta\rho$) and resolution ellipses.
- Compare inversion results with misfit contours; highlight underdetermined directions.

170–180 min: Wrap-up

- Key takeaways: forward/inverse distinctions, misfit and weighting, non-linearity, non-uniqueness, damping.
- Open Q&A; preview of next week's material.

Materials

- Linear/non-linear handout or slide deck.

- Activity D handouts (gravity and magnetic curves-only plots).
- Buried sphere app (laptop per group).
- Whiteboard or shared notepad for Jacobian derivations.

Assessment

- Activity notes (classification & argumentation).
- Jacobian derivation steps.
- Short reflection or exit-ticket on non-linearity and non-uniqueness.

References (suggested for students)

Further Reading:

[1] R. Blakely, *Potential Theory in Gravity and Magnetic Applications*, Cambridge University Press, 1995.
[2] W. M. Telford, L. P. Geldart, and R. E. Sheriff, *Applied Geophysics*, 2nd ed., Cambridge University Press, 1990.
[3] W. Menke, *Geophysical Data Analysis: Discrete Inverse Theory*, 3rd ed., Academic Press, 2012.
[4] R. C. Aster, B. Borchers, and C. H. Thurber, *Parameter Estimation and Inverse Problems*, 3rd ed., Elsevier, 2019.
[5] A. Tarantola, *Inverse Problem Theory and Methods for Model Parameter Estimation*, SIAM, 2005.

Activity A
Linear or Non-linear?

Goal. For each of the following forward relations drawn from potential fields, geothermics, and physical property relations, decide whether it is (i) linear, (ii) non-linear, or (iii) non-linear but can be linearized. If non-linear, describe how you would linearize or re-parameterize it. Justify your choice with respect to the model parameters that would be inverted.

Inversion problems can be cast as $\mathbb{A}\mathbf{m} = \mathbf{d}$ for linear problems and $A(m) = d$ for non-linear problems. Where d is the data and A is the operator or design (more on that below). The difference may appear subtle, though in reality the difference is that the linear equation can be written as a linear algebra problem as the model parameters are independent and discrete. In the non-linear formulation, the one or more model parameters are dependent upon another. As a result they cannot be fully separated unless they can be linearized.

Let's explore what A, m, and d are a bit more intuitively. The data, d, are the results of an experiment. Identifying the data are generally straightforward. The operator and model parameters can takes a bit more getting used to. The model parameters, m, are the physical or virtual properties of the system and initially unknown. Once you determine these properties, the forward model becomes a prediction. The operator or design matrix, A, can be thought of *as the experiment itself*, and is dependent upon the geometry, conditions of the experiment, and the physics. The model tells us what exists, the operator tells us how we observe it.

When we are doing a geophysical experiment with the Earth as the medium,

- **m** → the model is Earth, what it is made of and its physical properties. Changing **m** changes the Earth.
- **A** → how Earth talks to the instrument, how the Earth is sampled by the experiment, i.e., the rules of the experiment. Changing **A** changes the experiment, i.e., same Earth + different experiment → different data.
- **d** → the data is the outcome of the experiment, what we measure

In most geophysical problems, **A** is

- gravity – distances from mass elements
- magnetics – sensor postions and dip angle
- heat flow – layer thicknesses and boundary conditions
- electromagnetics – distances from conductive/resistive bodies
- seismology – ray paths or travel times

So the entries in **A** are determined by:

- positions
- distances
- times
- physical laws
- boundary conditions

Put simply, the matrix $\mathbb{A}$ encodes the physics and geometry of how the experiment samples the model, **m**, resulting in the observations, **d**.

Instructions (15–20 min)

1. Work in groups of 3–4.
2. Identify the data and the experimental quantities.
3. For each equation, identify the quantity or quantities that would be treated as the *model parameter(s)* in an inversion.

4. Classify the mapping data → model as linear or non-linear in those parameters.
5. If non-linear, propose a linearization (e.g., log-transform, Taylor expansion/Jacobian around p_0, or a change of variables) and note any loss of identifiability.

Equations

Q1. **Gravity (Bouguer slab):** We'll see this equation again as it is used to model the gravity due to an infinite slab.

$$\Delta g = 2\pi G(\rho_s - \rho_c)H$$

Q2. **Gravity (buried sphere):** This gravity solution is one of the simplest models for a gravity anomaly. We will explore it further.

$$g(r) = \tfrac{4}{3}\pi a^3 G(\rho_s - \rho_c)\,\frac{z}{(r^2+z^2)^{3/2}}$$

Q3. **Thermal conductivity $k(T,X)$:** We'll use this formulation in the problem set. This equation is used to estiamte thermal conductivity model for a solid solution series for a mineral.

$$k(T,X) = (a + bX + cX^2)\,(298/T)^n$$

Q4. **Thermal conductivity $k(T,P)$:** This form of thermal conductivity is often used to estimate thermal conductivity for a rock.

$$k(T,P) = \frac{k_0}{A+BT}\,(1+cP)$$

Q5. **Isostasy (Airy):** Airy isostasy is an end-member isostatic model, used to explain topographic variations due to differences in crustal thickness.

$$\Delta t = \frac{\rho_c}{\rho_m - \rho_c}\,h$$

Q6. **Isostatic gravity correction (column):** Gravity can be corrected for a mass deficit created by variations in crustal thickness via an isostatic correction.

$$\Delta g_{\text{iso}} \approx 2\pi G(\rho_m - \rho_c)\,\Delta t - 2\pi G\,\rho_c\,h$$

Q7. **Magnetics (2-D infinite dike):** The ocean floor records chaotic reversals in Earth's magnetic field. The remnant magnetic field is "frozen" in dikes produed at the mid-ocean ridge.

$$\Delta T(x) = C\left[\arctan\left(\frac{x - x_1}{z}\right) - \arctan\left(\frac{x - x_2}{z}\right)\right]$$

Q8. **Layered 1-D heat with sources:** Geotherms can largely be explained by 1-D heat flow through a layered Earth. We'll see this again when we start geothermics.

$$T_i(z) = T_{i-1} + \frac{q_{i-1}}{k_i}(z_i - z_{i-1}) - \frac{A_i}{2k_i}(z_i - z_{i-1})^2$$

Q9. **Density–porosity mix:** Sedimentary rocks include pore space (porosity) that may is filled with a fluid (air, water, brine, gas, oil, etc.).

$$\rho_{bulk} = (1-\phi)\,\rho_{matrix} + \phi\,\rho_{fluid}$$

Q10. **Gardner's relation (density–velocity):** Many solutions in geophysics involve trying to estimate one property from another, in this case density from seismic velocity.

$$\rho = a\,v^b$$

Q11. **Gravity (discrete 2-D prisms):** While we can produce analytic equations for a variety of idealized geologic bodies with simple geometries (e.g., buried spheres as above), to produce more geologically realistic bodies we can produce structures from arbitrarily-shaped polygons.

$$g_z = 2G\Delta\rho \sum_{i=1}^{N}\left[(x_{i+1} - x_i)\log\left(\frac{r_{i+1}}{r_i}\right) + (z_{i+1} - z_i)\arctan\left(\frac{x_i z_{i+1} - x_{i+1} z_i}{x_i x_{i+1} + z_i z_{i+1}}\right)\right]$$

where

$$r_i = \sqrt{x_i^2 + z_i^2}, \qquad (x_{N+1}, z_{N+1}) = (x_1, z_1).$$